

PaperToSkill: Turning Research Papers into Portable Agent Skills

Anonymous submission

Abstract

Many LLM and agent papers introduce workflows that could help ordinary users and agent builders, but those workflows are usually described for human reading rather than direct reuse. Summaries compress the idea, but often discard the procedural details, validation checks, assumptions, and failure branches needed to apply the method. We introduce PaperToSkill, a paper-to-skill conversion workflow that turns source-anchored paper notes into compact, human-editable agent skills. Each skill records the paper’s contribution, use conditions, workflow, validation checks, failure cases, transfer notes, and source anchors. In a four-paper benchmark over AI Scientist-v2, Reflexion, AIDE, and Toolformer, PaperToSkill generates skills that score 20/20 on deterministic structural rubrics, preserve more deterministic operational coverage than generic-summary and abstract-only baselines, remain under a 1200-word compactness budget, and substantially reduce deterministic input-token proxy size relative to full extracted papers. These results support PaperToSkill as a reproducible conversion layer from curated paper notes to portable skills. All four live-transfer saved-response sets are collected and scored under a deterministic output-contract evaluator; these saved response files are not human semantic fidelity or downstream execution-outcome evidence.

Introduction

Using LLMs is becoming a practical skill for researchers, builders, and non-expert users. At the same time, many of the best ways to use LLM agents are introduced in papers whose methods are difficult to operationalize. A paper may describe a search policy, memory loop, validation stage, or interface contract, but a user who wants to reuse the contribution must translate prose into an agent workflow by hand.

PaperToSkill studies a simple hypothesis: papers can be converted into compact, editable skills that preserve enough procedural knowledge for agents and users to reuse the main contribution. We use “skill” to mean a natural-language operational artifact with an entry-point `SKILL.md`, similar to formats used by agent harnesses that load task-specific instructions. Unlike a summary, a skill is not only explanatory. It should specify when to use the method, which inputs are

required, which steps to execute, how to validate progress, which failures to watch for, and how to adapt the instructions across harnesses.

This paper takes an artifact-first view. PaperToSkill is not presented as a fully automatic PDF understanding system. The current implementation converts curated, source-anchored paper notes into skills and source maps. This narrow scope is intentional: it lets us evaluate whether the conversion layer preserves operational knowledge before claiming end-to-end PDF automation.

Our current contributions are: (1) a paper-to-skill schema for operationalizing research contributions; (2) a deterministic extraction scaffold that emits `SKILL.md` plus source maps; (3) deterministic extracted-text-to-note scaffolds evaluated on Toolformer and AIDE; (4) a four-paper benchmark using AI Scientist-v2, Reflexion, AIDE, and Toolformer; and (5) deterministic/offline evaluations for structure, context coverage, compactness, source grounding, and transfer readiness.

Related Work

PaperToSkill is motivated by automated research and agent workflow systems. AI Scientist-v2 uses agentic tree search and staged experiment management for automated scientific discovery (Lu et al. 2025). AIDE frames ML engineering as search over code solutions (AIDE). Reflexion stores verbal feedback in episodic memory to improve later attempts without weight updates (Shinn et al. 2023). Toolformer shows how language models can generate and filter tool-use examples for API calls (Schick et al. 2023). Skill-library and interface work such as Voyager and SWE-agent also demonstrate that reusable skills and agent-facing interfaces can determine whether a method transfers beyond its original implementation (Wang et al. 2023; Yang et al. 2024).

These papers each contribute reusable procedures, but the procedures remain inside the paper or the original system. PaperToSkill targets the conversion layer: extracting operational knowledge from papers into compact skills with validation checks, failure branches, and transfer notes.

Method

PaperToSkill represents each converted paper as a skill package. The main file is `SKILL.md`; optional references include

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

a source map that links generated skill bullets to source note sections and line spans. The schema contains identity, central contribution, use conditions, required artifacts, workflow steps, validation checks, failure cases, transfer notes, and source notes that distinguish source-backed material from inferred guidance.

The deterministic extractor reads a source-anchored paper note, normalizes Markdown sections and list items, collects candidates from abstract, methods, experiments, limitations, and transfer sections, then emits a compact skill. Candidate limits keep the artifact within a practical context budget while preserving method, validation, and failure coverage. The extractor also writes a `references/source_map.json` file so later audits can inspect which source section supports each generated instruction.

The text-to-note scaffold is an earlier preprocessing step. It operates on newline-numbered extracted text, detects broad paper regions, selects line windows for methods, experiments, and limitations, and preserves line anchors. For two-column extracted text, it keeps raw line spacing and selects the keyword-bearing column while retaining the original source line range. The output is explicitly marked as an audit scaffold.

Experimental Setup

We evaluate four real agent-method papers: AI Scientist-v2, Reflexion, AIDE, and Toolformer. For each paper, we create a curated source-anchored note from the extracted paper text, generate a PaperToSkill skill, and compare it against two context baselines: a generic summary and an abstract-only context. For transfer ablation, we compare the full skill against the same skill with `Transfer Notes` removed and against the generic summary.

The metrics are deterministic: skill rubric score, context coverage, source-span validation, offline transfer-readiness, word count under a 1200-word compactness budget, character-based token/cost proxy, and a local tokenizer-aware input proxy using `o200k_base`. For saved model-ablation responses, we additionally compute a local output-token proxy. These metrics are reproducible gates. They do not replace live agent execution, provider billing, live invoices, success-per-dollar accounting, or completed human fidelity annotation. A billing evidence template and summary validator are prepared for future invoice or usage-export rows, but the current billing summary has 6 pending rows and 0 measured provider bills. We also prepare model-ablation prompt packets for Claude Opus 4.8, a GPT-family slot requested as GPT 5.5, and a later DeepSeek slot, and include a runner and response evaluator. Claude Opus 4.8 and GPT-family rows now have saved and scored responses for the current two-case protocol; DeepSeek remains pending user configuration. Provider/model availability failures are logged separately from model-quality results.

Results

All four generated skills score 20/20 on deterministic skill rubrics. On context coverage, the generated skills score 7.867/9 for AI Scientist-v2, 8.267/9 for Reflexion, 9.1/10

for AIDE, and 8.9/10 for Toolformer. The generic-summary and abstract-only baselines score substantially lower across all four cases (Table 1).

The generated skills remain compact: 782 words for AI Scientist-v2, 479 words for Reflexion, 927 words for AIDE, and 943 words for Toolformer, all under the 1200-word budget. Source validation finds support rates of 0.938, 1.0, 1.0, and 1.0 with zero invalid line ranges. The one weak AI Scientist-v2 span is recorded as a boundary case rather than treated as fully verified.

The tokenizer-aware context proxy shows the same compactness pattern relative to full paper contexts. Under `o200k_base`, generated skills use between 2.39% and 9.65% of the input tokens required by full extracted paper text (Table 3). A separate character-proxy report preserves the earlier $\lceil \text{characters}/4 \rceil$ estimate for reproducibility. We interpret both as coverage-preserving compression relative to full-paper context, not as provider billing evidence or a claim that skills are the shortest possible context.

Transfer-note ablations show a consistent offline readiness pattern. Full skills score 10/10 on the readiness metric across all four curated papers. Removing `Transfer Notes` lowers readiness to 7.6/10 in all four cases, while generic summaries score between 1.2/10 and 2.25/10. This supports the narrower claim that transfer notes encode artifact-readiness signals. It does not yet prove improved live cross-harness success.

The live-transfer saved-response evaluation covers all four paper packets. AI Scientist-v2, Reflexion, AIDE, and Toolformer each have six saved responses across Codex-style and Claude-style harness prompts and three context variants. The aggregate evaluator reports 24 total rows, 24 scored rows, 0 pending rows, and average normalized score 1.0. AI Scientist-v2, Reflexion, and AIDE rows score 11/11 each; Toolformer rows score 9/9 each. The AIDE run includes one provider fallback row where `claude-opus-4-8` closed the connection and `claude-opus-4-7` succeeded. These are deterministic output-contract scores over saved response files, not human-rated semantic fidelity or downstream execution-outcome rates.

The automatic note scaffold is evaluated separately on Toolformer and AIDE. The Toolformer auto-note-derived skill scores 20/20 on the deterministic rubric, 9.3/10 on context coverage, 10/10 on offline transfer readiness, and 1.0 source-span support with zero invalid ranges. The AIDE auto-note-derived skill scores 20/20, 8.467/10 on context coverage, 9.5/10 on offline transfer readiness, and 1.0 source-span support with zero invalid ranges (Table 4). These results support the narrower claim that extracted text can seed auditable note scaffolds. The local pipeline also includes a direct-PDF smoke path through `pdftotext -layout` that records the generated text source in the manifest, but this does not establish robust arbitrary-PDF automation.

Usage Examples and Pending Model Ablations

The repository includes usage examples for three practical workflows: loading a generated skill in a Codex-style harness, generating an automatic note scaffold from extracted

Table 1: Main deterministic/offline PaperToSkill results. Coverage scores are task-rubric scores; transfer readiness is an offline artifact-readiness metric.

Paper	Rubric	Skill	Summary	Abstract	Δ Sum.	Transfer	Support	Words
AI Scientist-v2	20/20	7.867/9	1.733/9	1.2/9	6.134	10/10	0.938	782
Reflexion	20/20	8.267/9	3.483/9	2.533/9	4.784	10/10	1.000	479
AIDE	20/20	9.1/10	1.916/10	1.333/10	7.184	10/10	1.000	927
Toolformer	20/20	8.9/10	2.5/10	1.534/10	6.400	10/10	1.000	943

Table 2: Transfer-note ablation. Removing only Transfer Notes lowers offline readiness across all curated skills.

Paper	Full	No Transfer
AI Scientist-v2	10/10	7.6/10
Reflexion	10/10	7.6/10
AIDE	10/10	7.6/10
Toolformer	10/10	7.6/10

text or a smoke-tested PDF source, and building model-ablation prompts. The model-ablation protocol produces a prompt grid for Claude Opus 4.8, a GPT-family slot requested as GPT 5.5, and a DeepSeek follow-up slot. The latest live recheck completed both Claude Opus 4.8 prompt rows, saved response files, and scored both rows 6/6 under the response rubric. A later GPT-family retry with the separate GPT credential profile saved both current GPT-family rows: the Toolformer row timed out on gpt-5.5 and then succeeded with gpt-5.4, while the AIDE row succeeded with gpt-5.5. Both GPT-family rows scored 6/6. Therefore, the paper treats Claude Opus 4.8 and GPT-family rows as completed for the current prompt protocol, while DeepSeek rows remain pending rather than negative model-quality evidence. A local output-token proxy over the four saved Claude/GPT-family responses counts 8,710 o200k_base output tokens: 2,272 for Claude Toolformer, 2,108 for Claude AIDE, 1,447 for GPT-family Toolformer, and 2,883 for GPT-family AIDE. This proxy is not provider billing, live-invoice, or success-per-dollar evidence.

Discussion

The main result is not that PaperToSkill “understands” papers in the broadest sense. The result is that a structured paper-note-to-skill pipeline can preserve operational knowledge that summaries discard. The skills include workflow steps, validation checks, failure cases, and source anchors, which are exactly the parts needed when a user wants an agent to apply a paper rather than merely explain it.

The benchmark also reveals useful failure modes. During the AIDE case, the initial extractor capped workflow, validation, and failure candidates too aggressively, which dropped data-preview and LLM-cost content. Earlier phases also exposed title parsing, source-audit mapping, and source-span line-counting bugs. These records are the type of failure branches PaperToSkill should preserve: not just final successes, but the ways extraction, evaluation, or external de-

pendencies can lose important operational content.

Limitations

The current work has several important limits. First, the main benchmark converts curated notes rather than arbitrary PDFs; automatic note scaffolds have only been retained on Toolformer and AIDE extracted text. Second, the metrics are deterministic and partly lexical, so they can over-credit exact matches or under-credit valid paraphrases. Third, live-transfer saved-response coverage is complete for the current prompt-packet protocol, but the scorer is deterministic and contract-based rather than a human semantic-fidelity or live execution-outcome evaluation. Fourth, human-fidelity packets, a reviewer handoff guide, a stricter blank annotation template, and a summary script are prepared, but no independent annotations have been completed. Fifth, compactness is measured by word count, deterministic character proxy, local tokenizer-aware input-token proxy, and local output-token proxy for saved model responses, not by provider-specific prices, live invoices, or live success per dollar.

Conclusion

PaperToSkill turns paper-derived procedural knowledge into portable, human-editable skills. In a four-paper benchmark, generated skills are compact, source-grounded, structurally valid, and more operationally complete than short summary baselines under deterministic evaluation. The next stage is to add human fidelity review, collect provider-specific billing or invoice evidence, evaluate downstream execution outcomes separately from saved-response scoring, add the user’s DeepSeek follow-up rows, and test papers whose contributions are less naturally procedural.

Table 3: Local tokenizer-aware context-size proxy using the o200k_base tokenizer. Counts are not provider bills, output-token costs, or success-per-dollar measurements.

Paper	Full Paper Tokens	Skill Tokens	Skill/Full	Reduction
AI Scientist-v2	45212	1079	0.0239	97.61%
Reflexion	16414	703	0.0428	95.72%
AIDE	13312	1285	0.0965	90.35%
Toolformer	20365	1255	0.0616	93.84%

Table 4: Automatic extracted-text note scaffolds are evaluated separately from curated-note cases.

Paper	Input	Rubric	Coverage	Transfer	Support	Words
Toolformer	Curated note	20/20	8.9/10	10/10	1.000	943
Toolformer	Auto note scaffold	20/20	9.3/10	10/10	1.000	1179
AIDE	Curated note	20/20	9.1/10	10/10	1.000	927
AIDE	Auto note scaffold	20/20	8.467/10	9.5/10	1.000	998

References

- AIDE. 2025. AIDE: AI-Driven Exploration in the Space of Code. arXiv:2502.13138.
- Lu, C.; Lu, C.; Lange, R. T.; Foerster, J.; Clune, J.; and Ha, D. 2025. The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search. arXiv:2504.08066.
- Schick, T.; Dwivedi-Yu, J.; Dessì, R.; Raileanu, R.; Lomeli, M.; Hambro, E.; Zettlemoyer, L.; Cancedda, N.; and Scialom, T. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. arXiv:2302.04761.
- Shinn, N.; Cassano, F.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. arXiv:2303.11366.
- Wang, G.; Xie, Y.; Jiang, Y.; Mandlkar, A.; Xiao, C.; Zhu, Y.; Fan, L.; and Anandkumar, A. 2023. Voyager: An Open-Ended Embodied Agent with Large Language Models. arXiv:2305.16291.
- Yang, J.; Jimenez, C. E.; Wettig, A.; Lieret, K.; Yao, S.; Narasimhan, K.; and Press, O. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. arXiv:2405.15793.